

Main Content

- Representation Methods of Discrete Signals
- Typical discrete signal

1. Representation Methods of Discrete Signals

$x(t)$ Evenly spaced T_s $x(nT_s)$, represented $x[n]$ $n = 0, \pm 1, \pm 2, \dots$

A **sequence** of function values, e.g. $\left\{ \dots 9, \underset{\substack{\uparrow \\ n=0}}{8}, 4, 1 \dots \right\}$

Waveform: The length of each line segment represents the magnitude of the corresponding sequence value

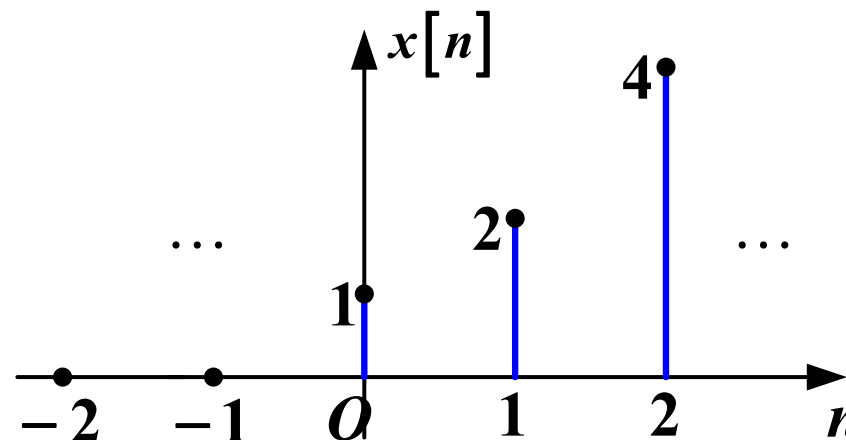
1. Representation Methods of Discrete Signals

$$x[n] = \begin{cases} 2^n, & n \geq 0 \\ 0, & n < 0 \end{cases}, \text{ write out its sequence form and plot the waveform}$$

Sequence form:

$$x[n] = \left\{ \dots, 0, 0, \underset{\substack{\uparrow \\ n=0}}{1}, 2, 4, 8, \dots \right\}$$

Waveform:

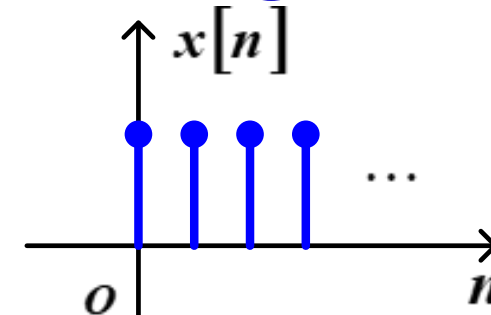


6.1 Typical Discrete-time signals

1. Representation Methods of Discrete Signals

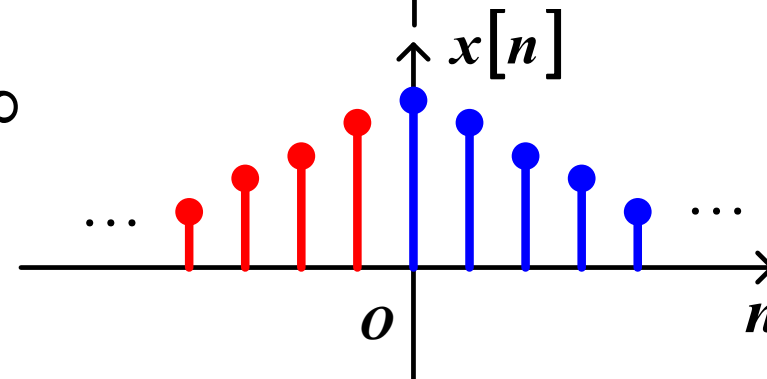
One-sided sequence

$$n \geq 0$$



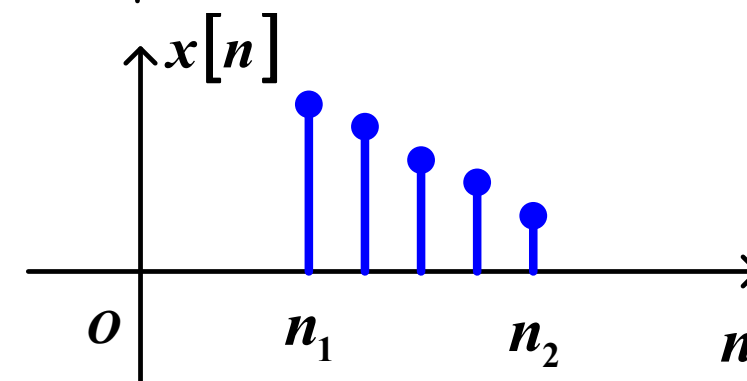
Two-sided sequence

$$-\infty \leq n \leq \infty$$



Finite-length sequence

$$n_1 \leq n \leq n_2$$



2. Typical discrete signal

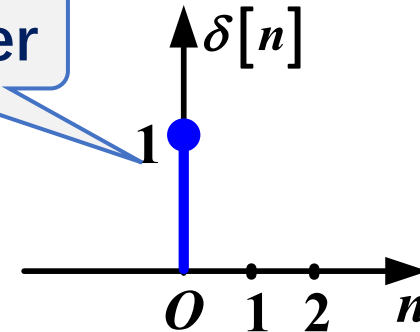
- Unit sample sequence (Discrete-time Impulse)
- Unit step sequence
- Rectangular sequence (Rectangular window)
- Ramp sequence
- One-sided Exponential Sequence
- Sinusoidal sequence
- Complex exponential sequence

6.1 Typical Discrete-time signals

(1) Unit sample sequence

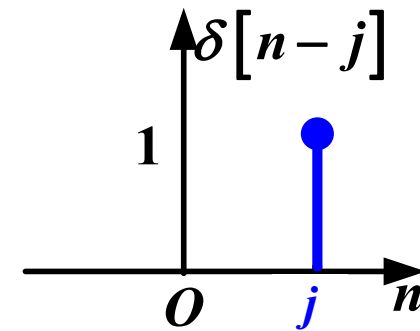
$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

Kronecker



Time-Shifting Property

$$\delta[n-j] = \begin{cases} 0, & n \neq j \\ 1, & n = j \end{cases}$$



Scaling Property

$$c\delta[n], c\delta[n-j]$$

Sampling Property

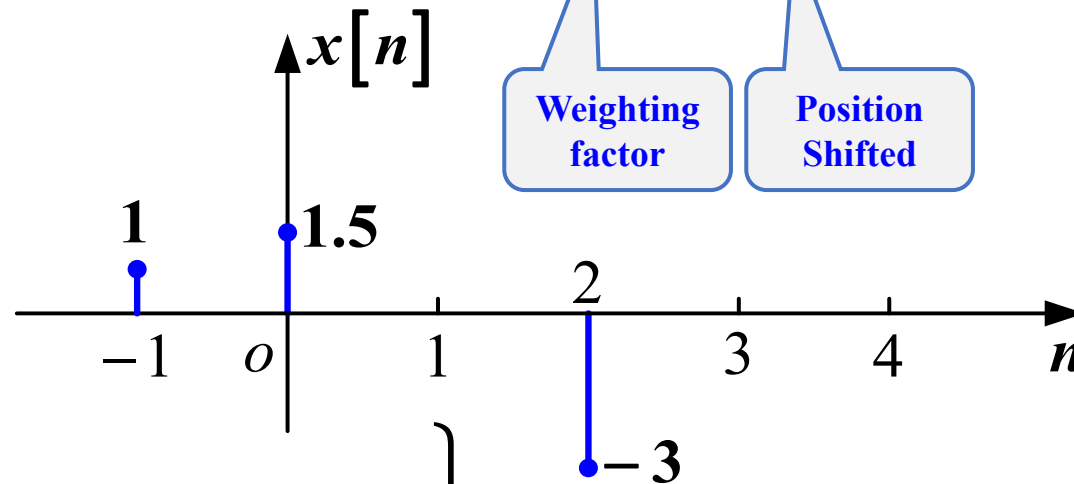
$$f[n]\delta[n] = f[0]\delta[n]$$

6.1 Typical Discrete-time signals

(1) Unit sample sequence

Representation of Arbitrary Sequences Using Unit Sample Signals

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n-m]$$

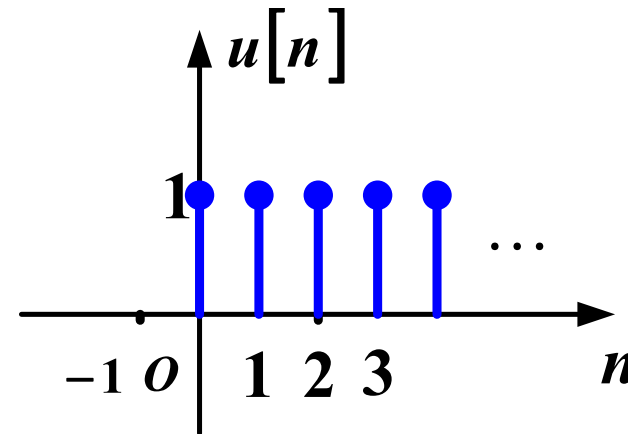


$$x[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1.5, 0, -3, 0, 0, \right\} = \delta[n+1] + 1.5\delta[n] - 3\delta[n-2]$$

6.1 Typical Discrete-time signals

(2) Unit step sequence

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



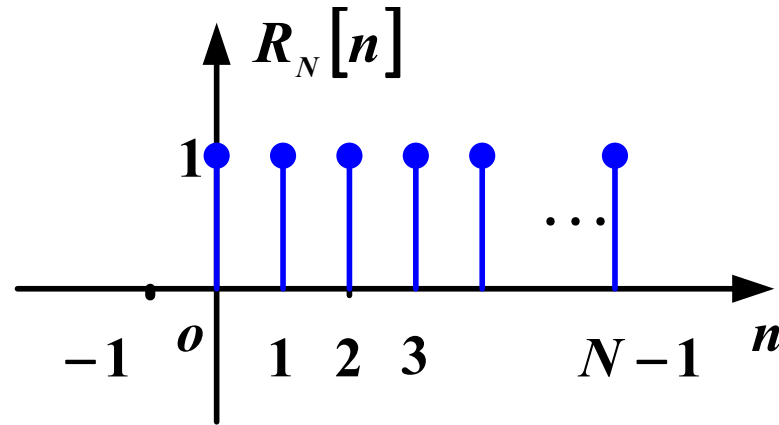
$u[n]$ can be expressed as an infinite sum of shifted unit sample signals

$$\begin{aligned} u[n] &= \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \dots \\ &= \sum_{k=0}^{\infty} \delta[n-k] \end{aligned}$$

$$\delta[n] = u[n] - u[n-1]$$

(3) Rectangular sequence

$$R_N[n] = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & n < 0, n \geq N \end{cases}$$



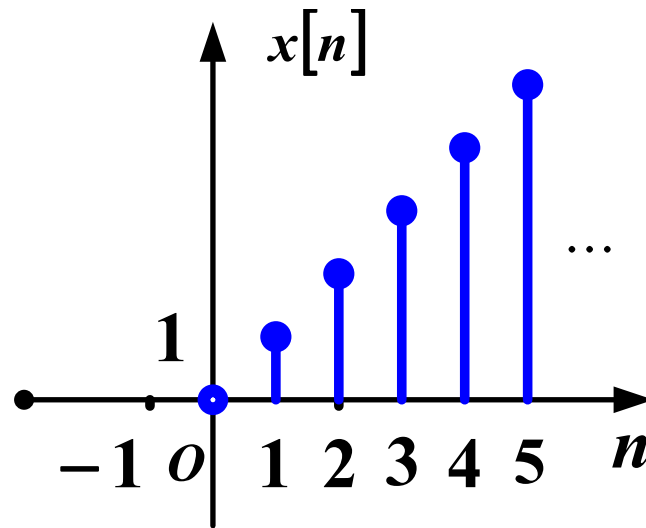
Relationship with $u[n]$: $R_N[n] = u[n] - u[n - N]$

$$R_3[n] = \delta[n] + \delta[n - 1] + \delta[n - 2]$$

6.1 Typical Discrete-time signals

(4) Ramp sequence

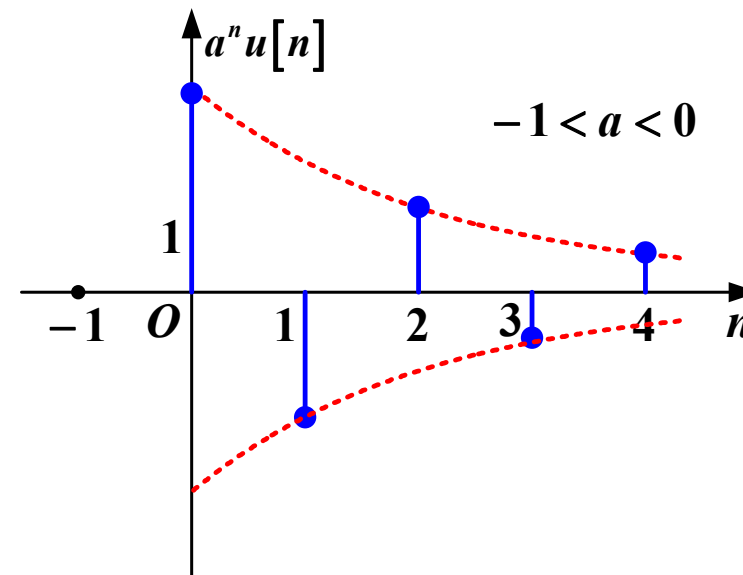
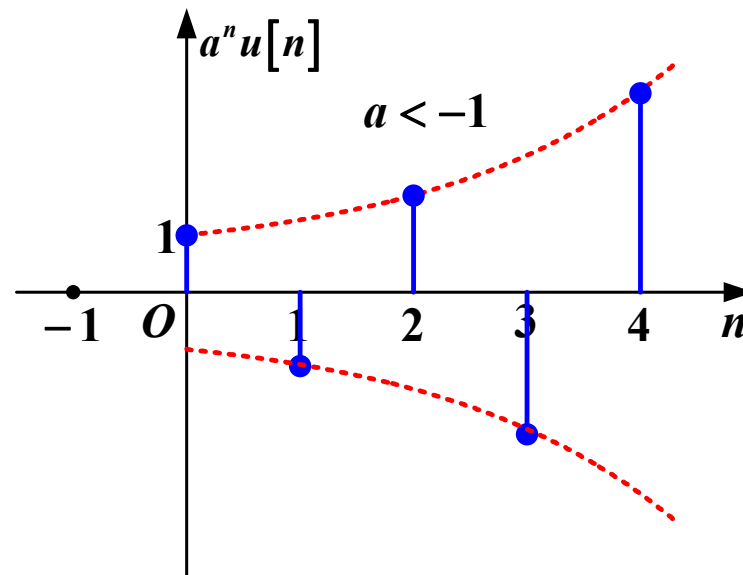
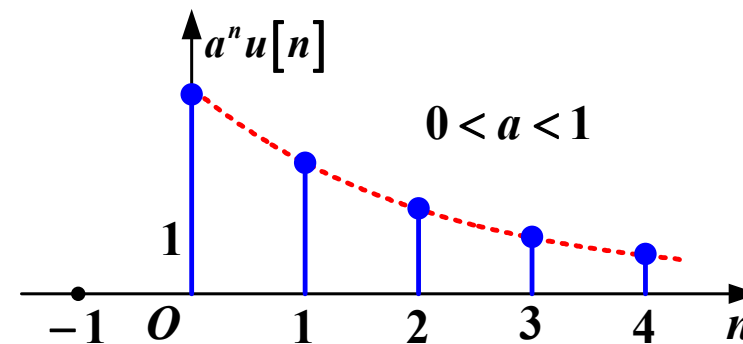
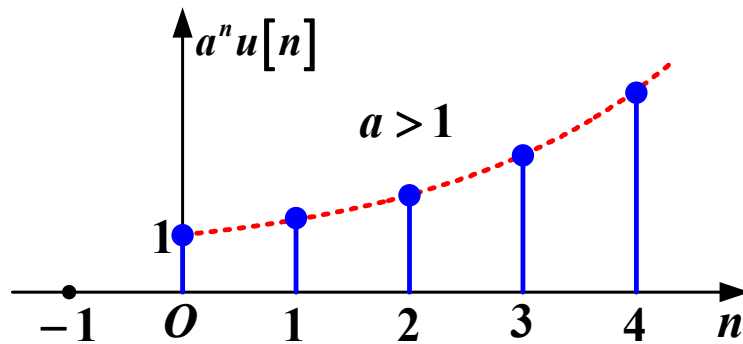
$$x[n] = nu[n]$$



6.1 Typical Discrete-time signals

(5) One-sided Exponential Sequence

$$x[n] = a^n u[n]$$



6.1 Typical Discrete-time signals

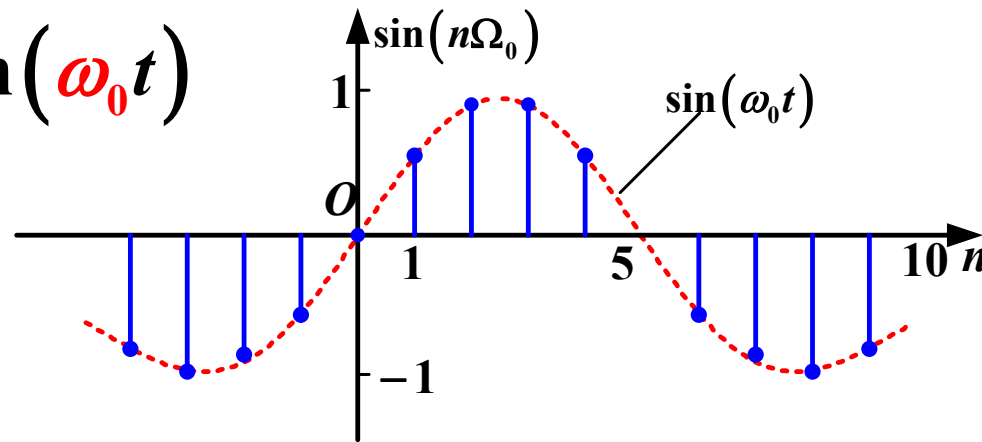
(6) Sinusoidal sequence

$$x_a(t) = \sin(2\pi f_0 t) = \sin(\omega_0 t)$$

Sampling interval T_s , Sampling frequency f_s

Sampled values at time instants $T_s n$

$$x_a(T_s n) = \sin(\omega_0 T_s n)$$

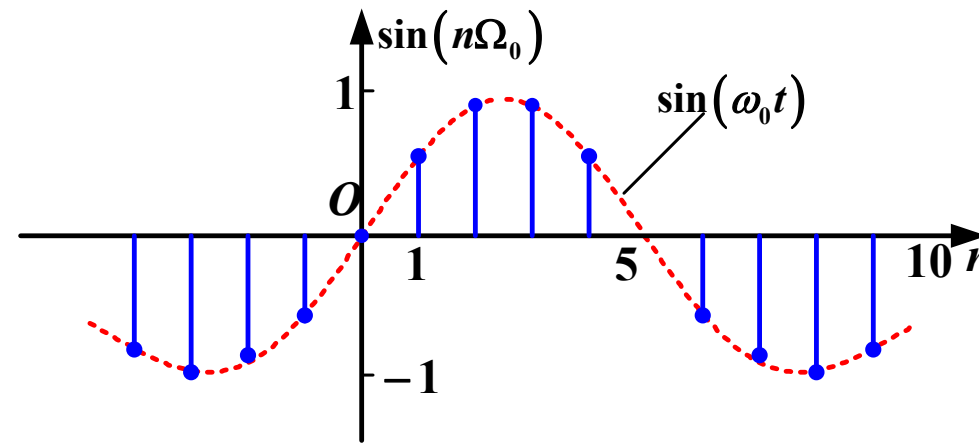


Represent the **sample values** of continuous time signals as sequences, with n as the variable

$$x[n] = \sin(\Omega_0 \cdot n)$$

Digital Angular Frequency $\Omega_0 = \omega_0 T_s = \frac{\omega_0}{f_s} = 2\pi \frac{f_0}{f_s}$ Unit: radians/sample

(6) Sinusoidal sequence



Periodicity of Sinusoidal Sequences

If a discrete sinusoidal sequence is **periodic**, it must satisfy

$$x[n] = \sin(\Omega_0 n)$$

$$x[n + N] = x[n]$$

N is called the **period** of the sequence and can be any **positive integer**

(6) Sinusoidal sequence

① $\frac{2\pi}{\Omega_0}$ is positive integer, then there is $N = \frac{2\pi}{\Omega_0}$

$$\sin[\Omega_0(n + N)] = \sin(n\Omega_0 + N\Omega_0)$$

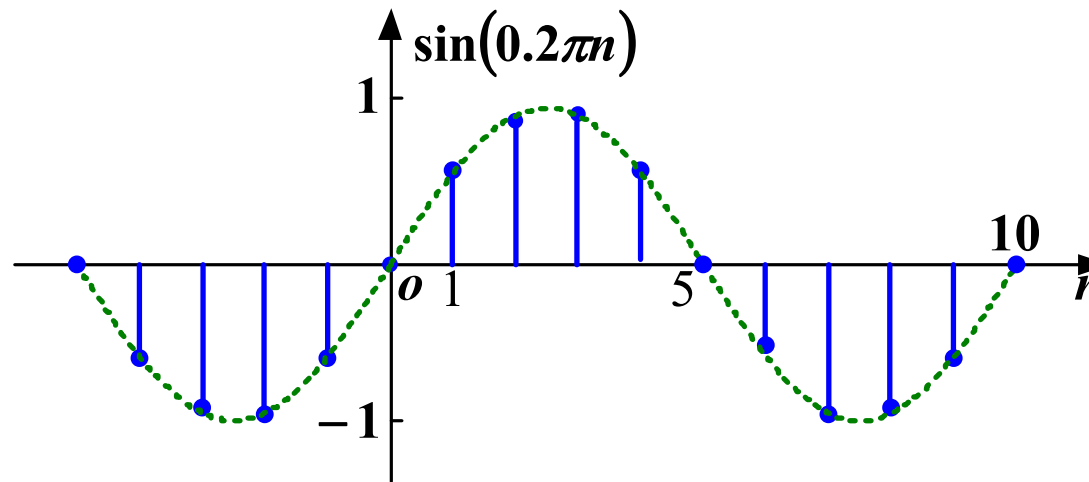
If $N\Omega_0 = 2\pi$, then $\sin[\Omega_0(n + N)] = \sin(\Omega_0 n)$

$$N = \frac{2\pi}{\Omega_0}$$

6.1 Typical Discrete-time signals

Example 1

$$x[n] = \sin(0.2\pi n)$$



$$\Omega_0 = 0.2\pi \quad \frac{2\pi}{\Omega_0} = 10 \quad \text{Period } N = 10$$

Repeats every **10** samples, with the sinusoidal envelope completing **1** full cycle

(6) Sinusoidal sequence

② $\frac{2\pi}{\Omega_0} = \frac{N}{m}$, N, m are positive integers, period $N = m \frac{2\pi}{\Omega_0}$

$$\sin[\Omega_0(n + N)] = \sin(n\Omega_0 + N\Omega_0)$$

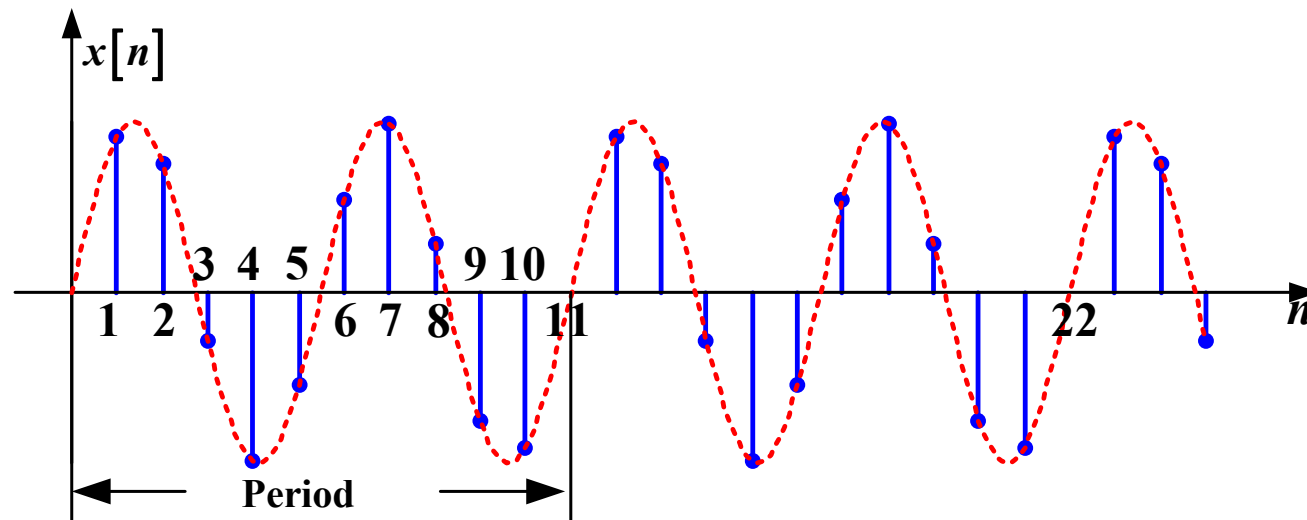
If $N\Omega_0 = m2\pi$, then $\sin[\Omega_0(n + N)] = \sin(\Omega_0 n)$

Period $N = m \frac{2\pi}{\Omega_0}$

6.1 Typical Discrete-time signals

Example 2

$$x[n] = \sin\left(\frac{4\pi}{11}n\right)$$



$$N = 11$$

$$\Omega_0 = \frac{4\pi}{11}, \quad \text{Then: } \frac{2\pi}{\Omega_0} = \frac{11}{2} = \frac{N}{m}$$

Repeats every **11** samples, with the sinusoidal envelope completing **2** full cycles

(6) Sinusoidal sequence

③ $\frac{2\pi}{\Omega_0}$ is irrational number, the sequence is aperiodic

$$\sin[\Omega_0(n+N)] = \sin(n\Omega_0 + N\Omega_0)$$

If $N\Omega_0 \neq m2\pi$, N , m are integers

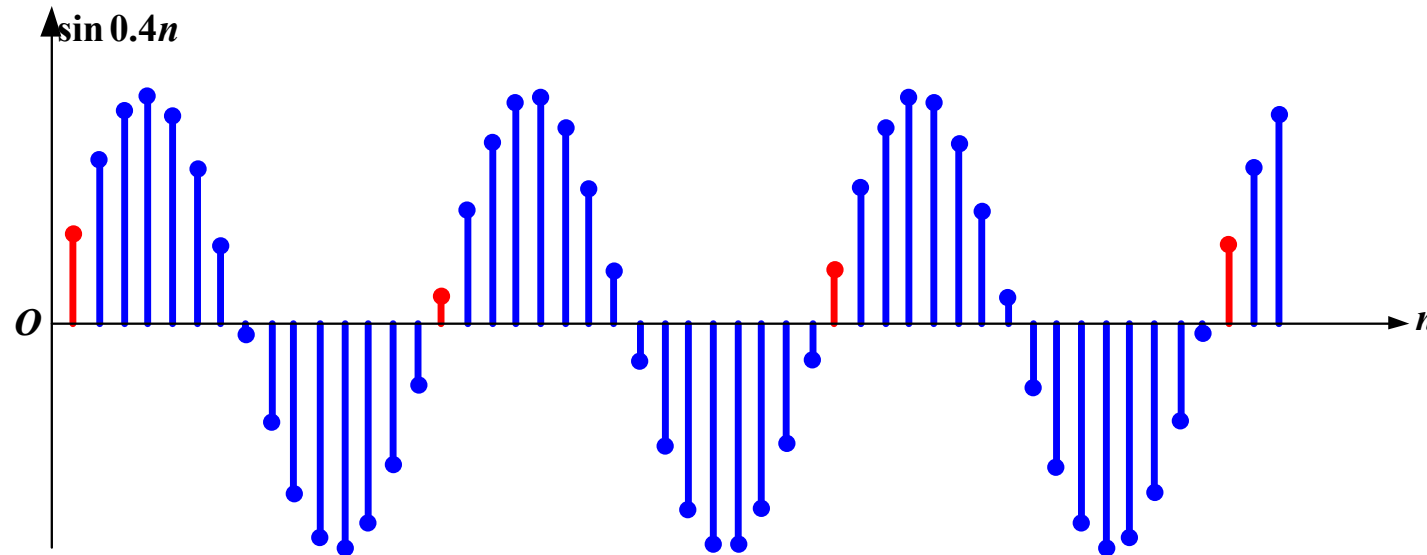
Then $\frac{2\pi}{\Omega_0}$ is irrational, aperiodic

6.1 Typical Discrete-time signals

Example 3

$$x[n] = \sin(0.4n)$$

Aperiodic
sequence



$$\Omega_0 = 0.4 \quad \frac{2\pi}{\Omega_0} = 5\pi \text{ irrational}$$

(7) Complex exponential sequence

$$x[n] = e^{j\Omega_0 n} = \cos \Omega_0 n + j \sin \Omega_0 n$$

Representing complex sequences in polar form:

$$x[n] = |x[n]| e^{j\arg[x[n]]}$$

$$|x[n]| = 1$$

$$\arg[x[n]] = \Omega_0 n$$

Summary

1. Representation Methods of Discrete Signals
2. Typical discrete signal
 - Unit sample sequence (Discrete-time Impulse)
 - Unit step sequence
 - Rectangular sequence (Rectangular window)
 - Ramp sequence
 - One-sided Exponential Sequence
 - Sinusoidal sequence
 - Complex exponential sequence

Main Content

- **Summation**
- **Multiplication**
- **Scaling**
- **Shifting (Time Shift)**
- **Reversal (Time Reversal)**
- **Difference (Discrete Derivative)**
- **Running sum (Cumulative Sum)**
- **Reshaping: Compression (Decimation), Extension (Interpolation)**
- **Energy of a sequence**
- **Power of a sequence**

6.2 Operations on discrete-time signals

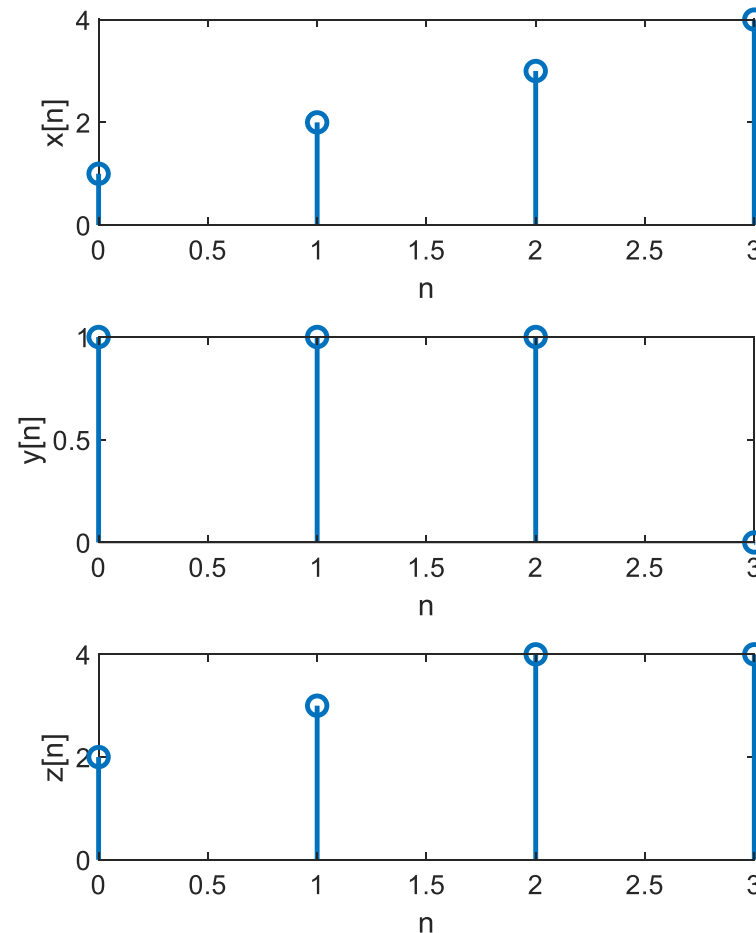
1. Summation

$$z[n] = x[n] + y[n]$$

Example 1:

$$x[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 3, 4 \right\}$$

$$y[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1, 1 \right\}$$



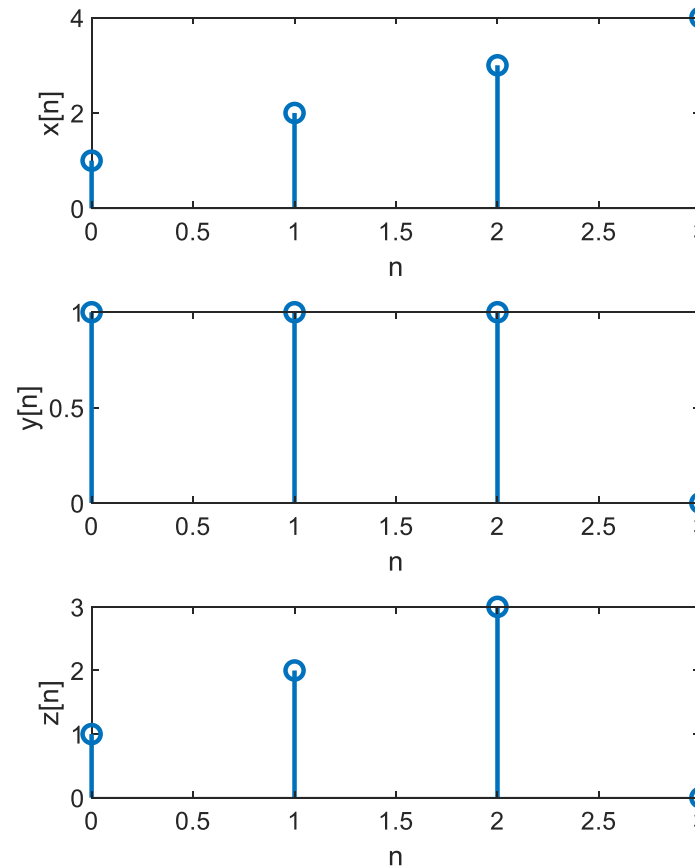
2. Multiplication

$$z[n] = x[n] \cdot y[n]$$

Example 2:

$$x[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 3, 4 \right\}$$

$$y[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1, 1 \right\}$$



3. Scaling

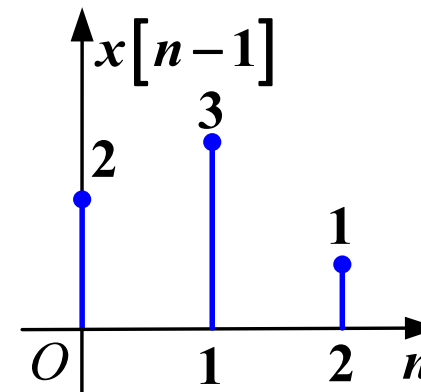
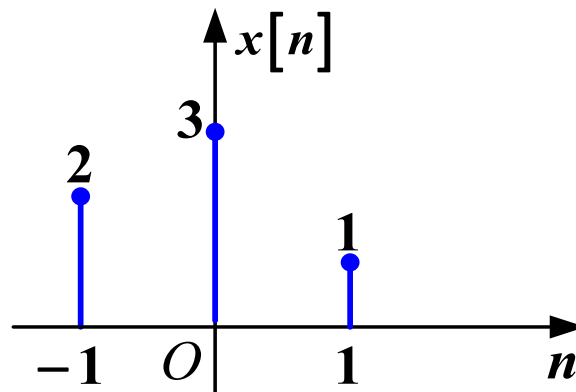
$$z[n] = ax[n]$$

4. Shifting

$$z[n] = x[n - m] \quad \text{Right Shift} \quad (m > 0)$$

$$z[n] = x[n + m] \quad \text{Left Shift} \quad (m > 0)$$

Example 3:



5. Reversal

$$z[n] = x[-n]$$

6. Difference

Forward difference: $\Delta x[n] = x[n+1] - x[n]$

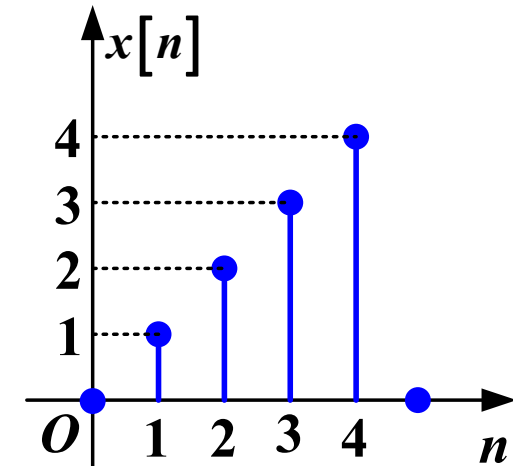
Backward difference: $\nabla x[n] = x[n] - x[n-1]$

7. Running sum

$$z[n] = \sum_{k=-\infty}^n x[k]$$

Example 4: calculate the running sum $y[n]$ of this signal

$n = 0$	$y[0] = 0$
$n = 1$	$y[1] = 0 + 1 = 1$
$n = 2$	$y[2] = 0 + 1 + 2 = 3$
$n = 3$	$y[3] = 0 + 1 + 2 + 3 = 6$
$n = 4$	$y[4] = 0 + 1 + 2 + 3 + 4 = 10$
$n > 4$	$y[n] = 10$



8. Reshaping

Decimation

$$x[n] \rightarrow x[Nn]$$

N is a positive integer

Interpolation

$$x[n] \rightarrow x\left[\frac{n}{N}\right], N \text{ is a positive integer}$$



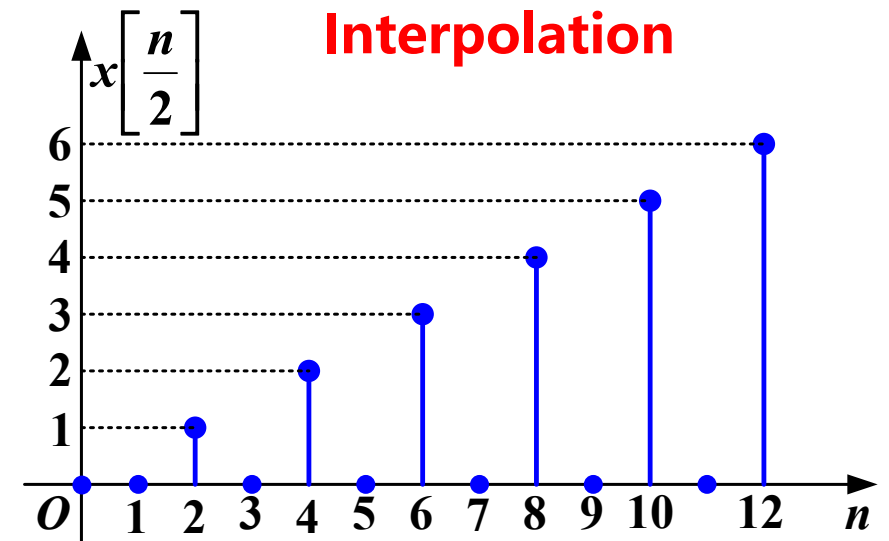
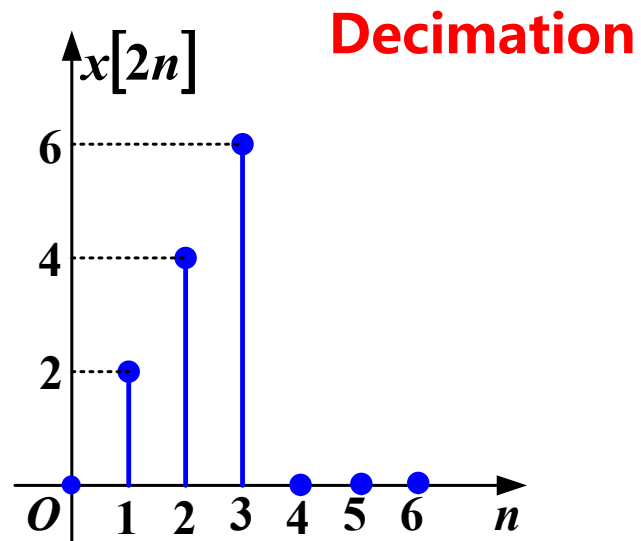
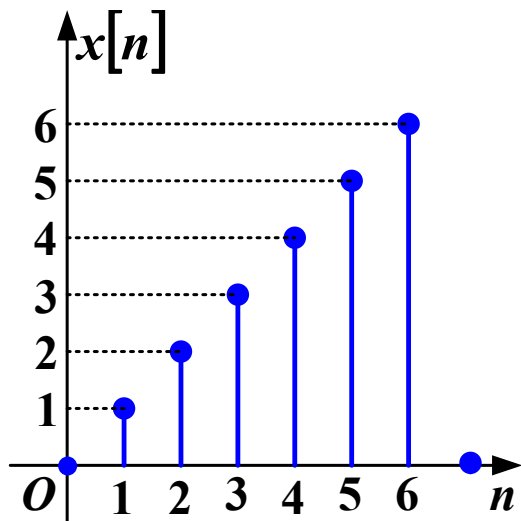
Decimation. Etching by William Hogarth in *Beaver's Roman Military Punishments* (1725)
[https://en.wikipedia.org/wiki/Decimation_\(Roman_army\)](https://en.wikipedia.org/wiki/Decimation_(Roman_army))

Note: Sometimes it is necessary to remove certain points or pad with corresponding zero values

6.2 Operations on discrete-time signals

Example 5

Given waveform of $x[n]$, please draw $x[2n]$, $x\left[\frac{n}{2}\right]$



9. Energy of a sequence

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The energy of a discrete-time sequence $x[n]$ over a finite interval $-K \leq n \leq K$ is defined as

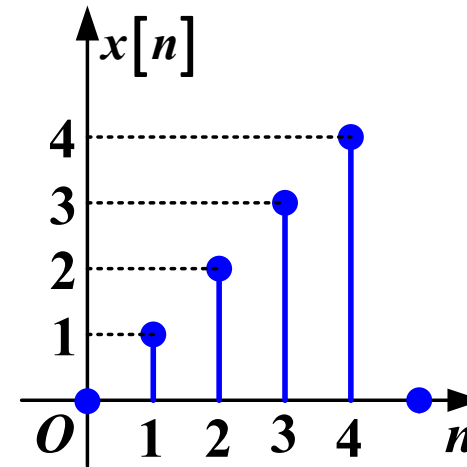
$$E_K = \sum_{n=-K}^K |x[n]|^2$$

6.2 Operations on discrete-time signals

Example 6

Calculate the energy E of this signal

$$\begin{aligned} E &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\ &= 1^2 + 2^2 + 3^2 + 4^2 \\ &= 30 \end{aligned}$$



10. Power of a sequence

Average power of aperiodic signals

$$P = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2$$

Average power of a periodic sequence with a period of N

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$$

Summary

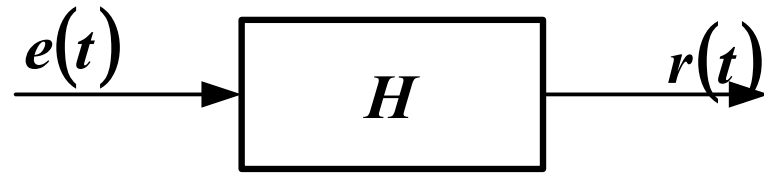
- **Summation**
- **Multiplication**
- **Scaling**
- **Shifting (Time Shift)**
- **Reversal (Time Reversal)**
- **Difference (Discrete Derivative)**
- **Running sum (Cumulative Sum)**
- **Reshaping: Compression (Decimation), Extension (Interpolation)**
- **Energy of a sequence**
- **Power of a sequence**

Main Content

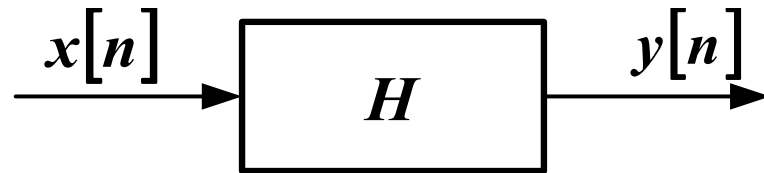
- Continuous-Time and Discrete-Time Systems
- Memoryless (Instantaneous) Systems vs. Dynamic Systems
- Lumped-Parameter Systems and Distributed-Parameter Systems
- Invertible Systems and Non-Invertible Systems
- Linear Systems
- Time-Invariant Systems
- Causal Systems
- Stable Systems

Key Focus: Study of lumped-parameter, linear, time-invariant (LTI) systems under deterministic signal excitation

1. Continuous-Time and Discrete-Time Systems



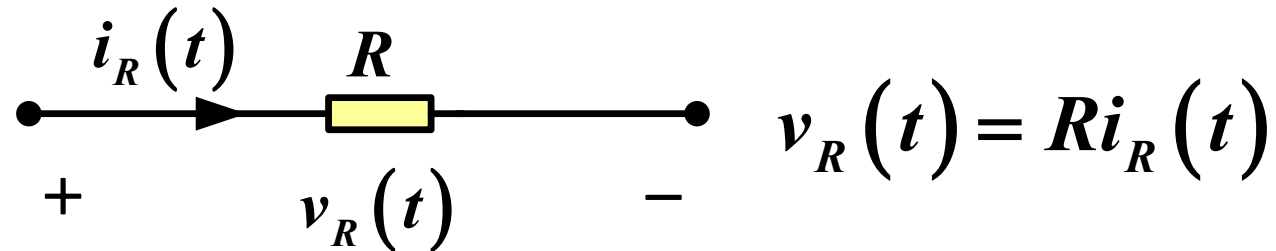
$$r(t) = H[e(t)]$$



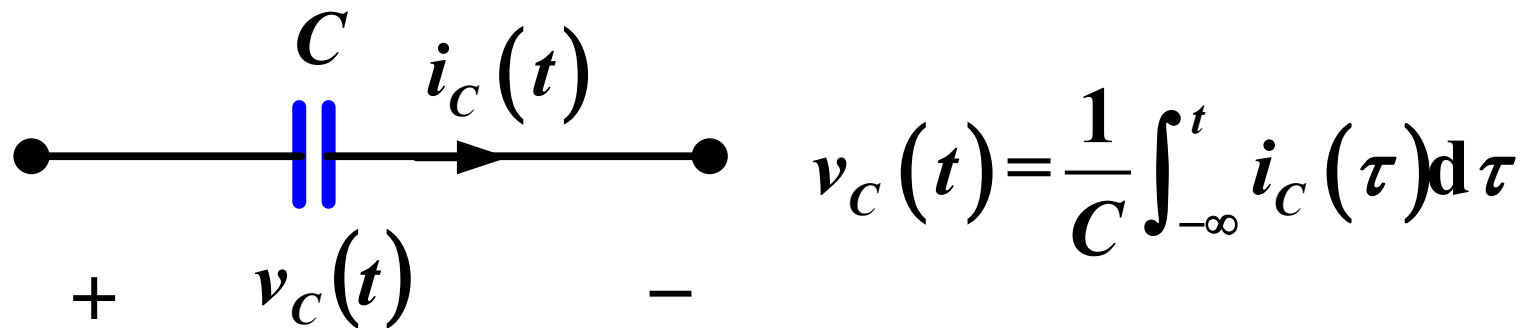
$$y[n] = H[x[n]]$$

2. Instantaneous Systems vs. Dynamic Systems

Instantaneous Systems (Memoryless): Algebraic Equation



Dynamic systems (systems with memory): differential or difference equations



3. Lumped-Parameter and Distributed-Parameter

Lumped parameter system: Ordinary differential equation
(*time – dependent only: t*)
Low-frequency electrical circuits (RLC)

Distributed Parameter Systems: Partial Differential Equations
(*time and space – dependent: t, x, y, z*)
Waveguides and transmission lines

4. Invertible Systems and Non-Invertible Systems

Reversible system: If the system produces different responses under different excitation signals, it is called a reversible system

Application: Utilizing inverse systems to achieve system compensation

$$r(t) = \eta |e(t)|^2 \quad (0 < \eta < 1)$$

Irreversible system: If different inputs produce identical outputs, making it impossible to determine the original input from the output alone

Characteristics of irreversible systems:

- The input signals $e(t)$ and $-e(t)$ produce the same output
- It is impossible to uniquely determine the input signal from the output
- The system is irreversible

5. Linear Systems

(1) Definition

Linear system: A system that exhibits linear properties

Linearity: Characterized by homogeneity and superposition

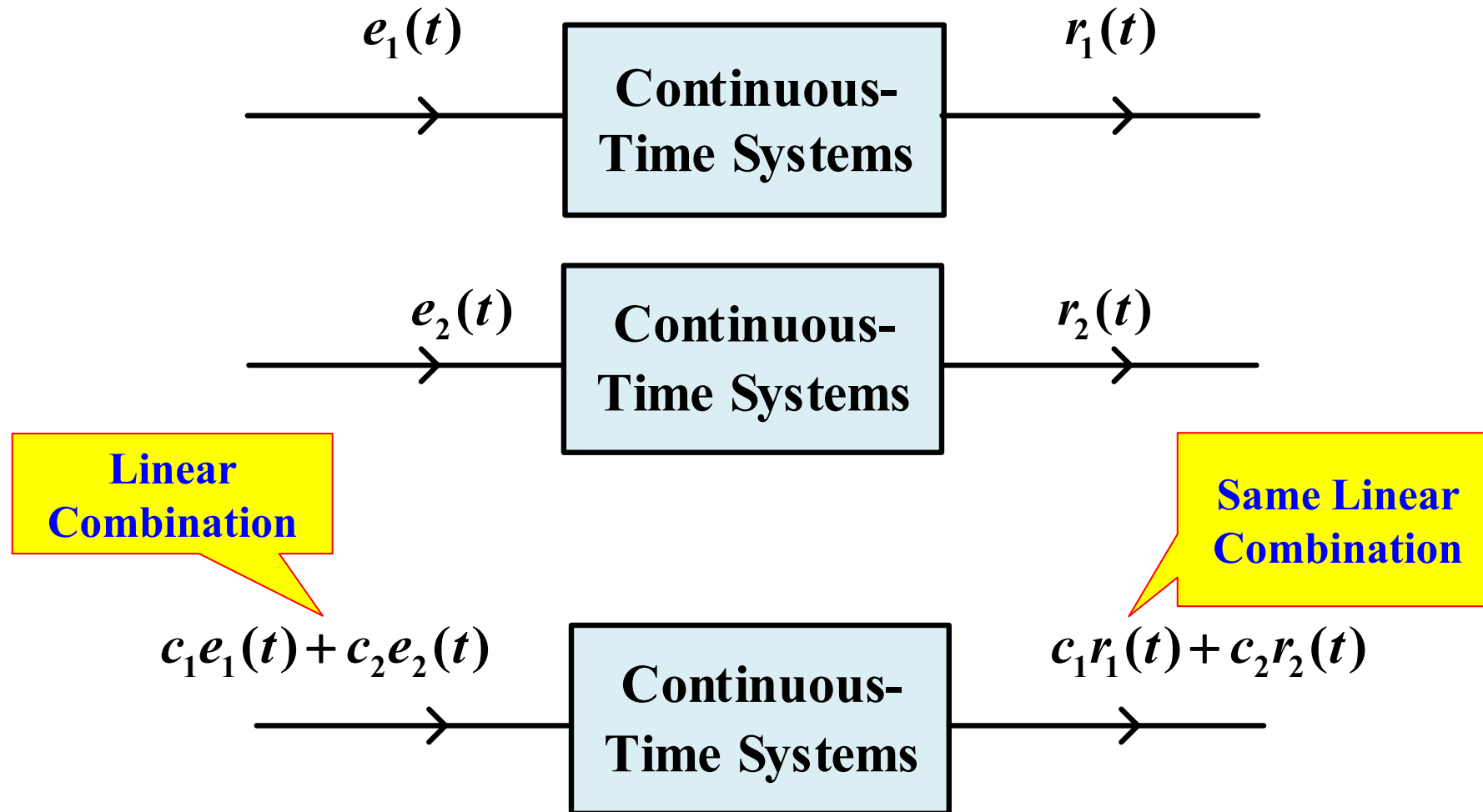
Homogeneity:

$$e(t) \rightarrow r(t) \Rightarrow Ke(t) \rightarrow Kr(t)$$

Superposition:

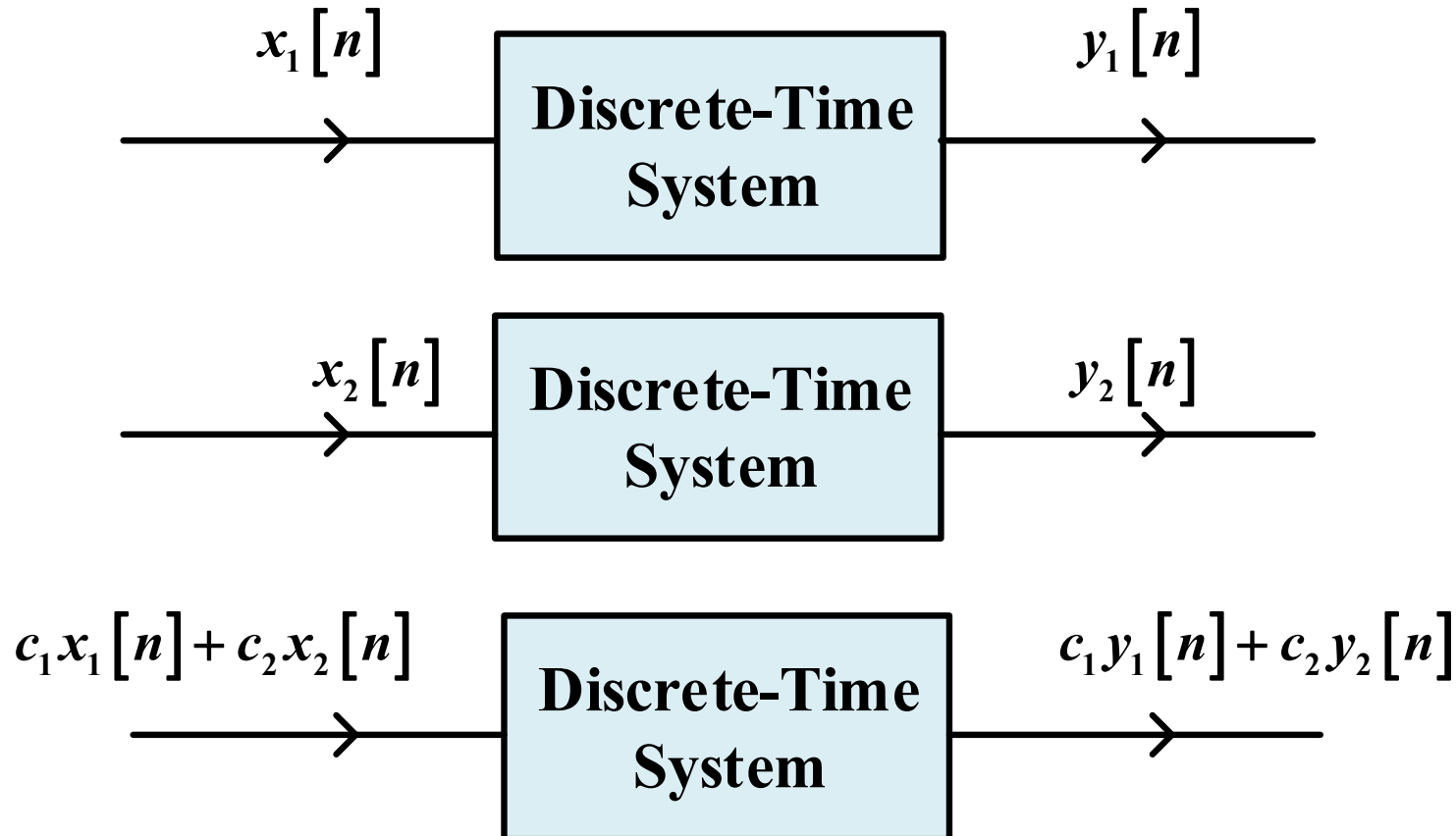
$$\left. \begin{array}{l} e_1(t) \rightarrow r_1(t) \\ e_2(t) \rightarrow r_2(t) \end{array} \right\} \Rightarrow e_1(t) + e_2(t) \rightarrow r_1(t) + r_2(t)$$

(2) Linear Properties of Continuous Systems



(3) Linear Properties of Discrete Systems

Linearity: Both homogeneity and superposition hold



(4) Method for Judging Linear Characteristics

First perform linear combination, then pass through system = First pass through system, then perform linear combination

$$\text{If } H[C_1 e_1(t) + C_2 e_2(t)] = C_1 H[e_1(t)] + C_2 H[e_2(t)]$$

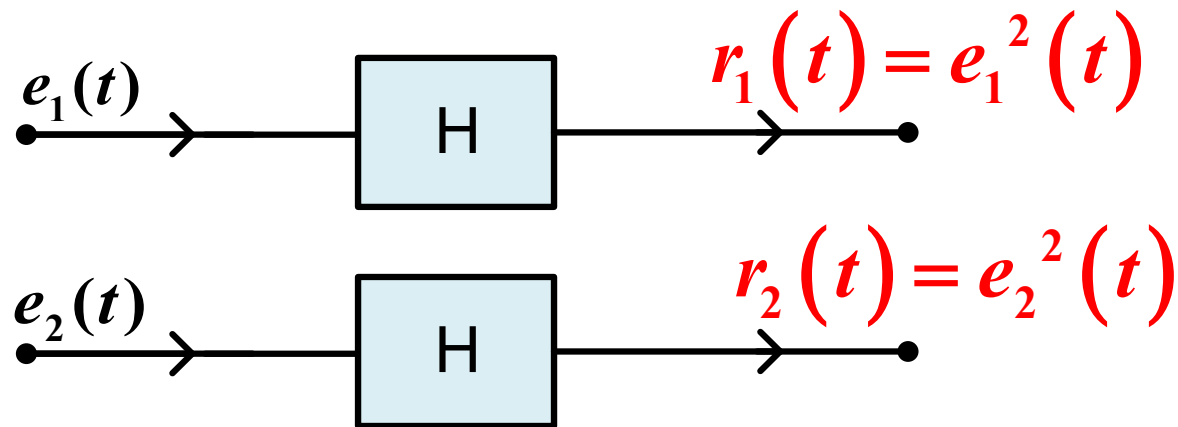
Then the system $H[\cdot]$ is a linear system, otherwise it is a nonlinear system

Note: The external excitation and non-zero state of the system must be handled separately

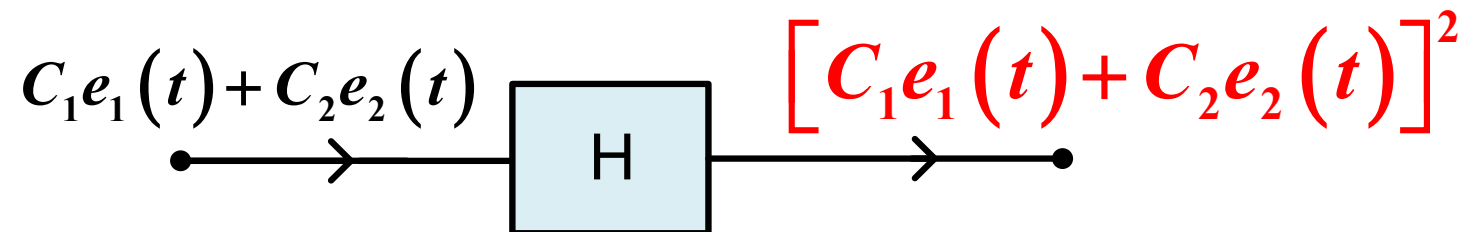
Example 1

Is the system $r(t) = e^2(t)$ a Linear system?

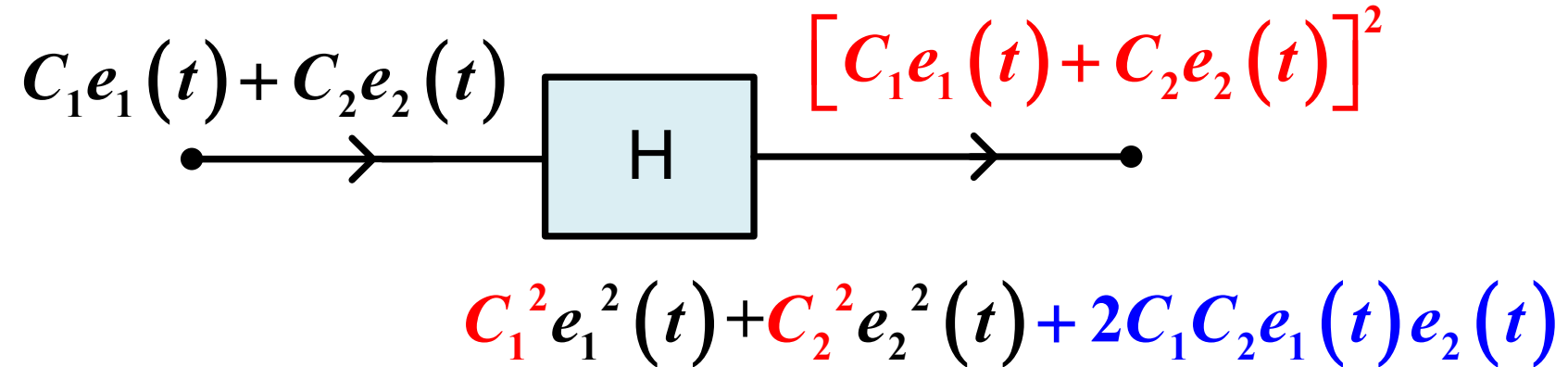
For any signals $e_1(t), e_2(t)$



Linearly combine the two signals and input into the system



Example 1



The linear combination of output signals $r_1(t)$ and $r_2(t)$ is

$$C_1 r_1(t) + C_2 r_2(t) = C_1 e_1^2(t) + C_2 e_2^2(t)$$

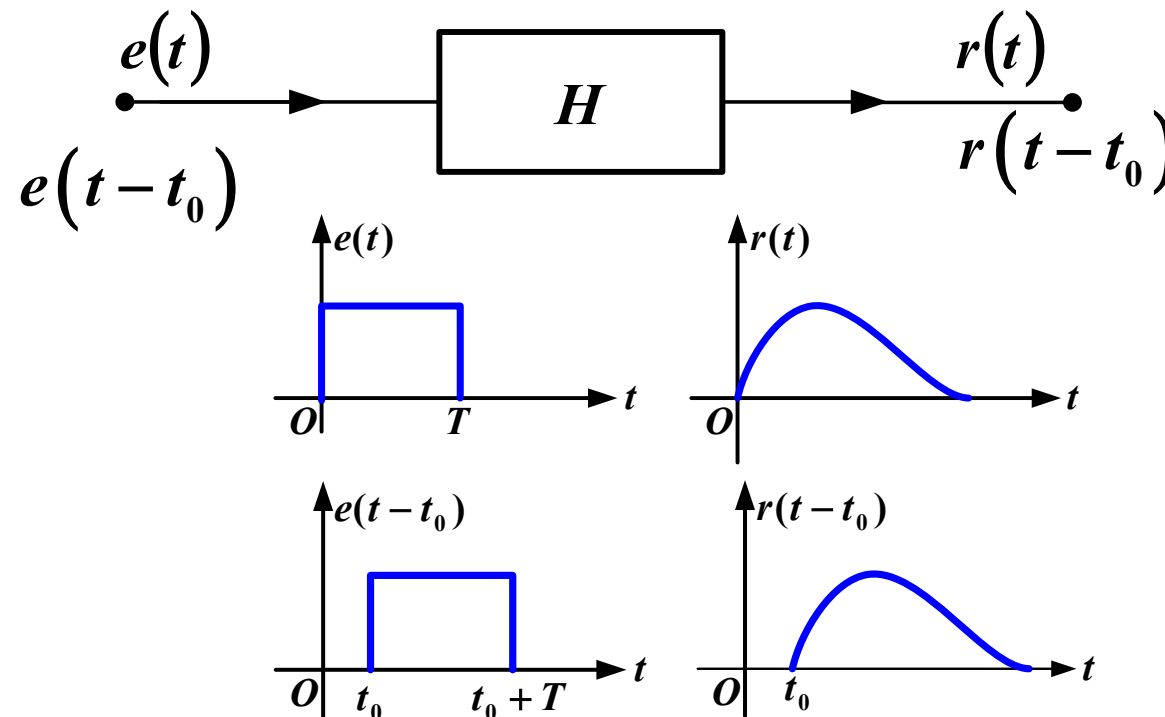
Obviously, the two results are different

Therefore, the system is **nonlinear**

6. Time-Invariant Systems

(1) Definition

A system is called time-invariant if, under **zero initial conditions**, its output signal is **independent of the time** at which the input signal is applied to the system; otherwise, it is called time-varying



(2) Time-Invariance of Continuous Systems

Explanation:

From a circuit analysis perspective:

Whether the **component parameters vary with time**

From an equation perspective:

Whether the **coefficients of the equation vary with time**

Judgment Method:

First time-shift, then pass through system = First pass through system, then time-shift

If $H[e(t - \tau)] = r(t - \tau)$

Then the system $H[\cdot]$ is time-invariant; otherwise, it is time-varying

Example 2

To judge whether the system $r(t) = \frac{de(t)}{dt}$ is a time invariant system

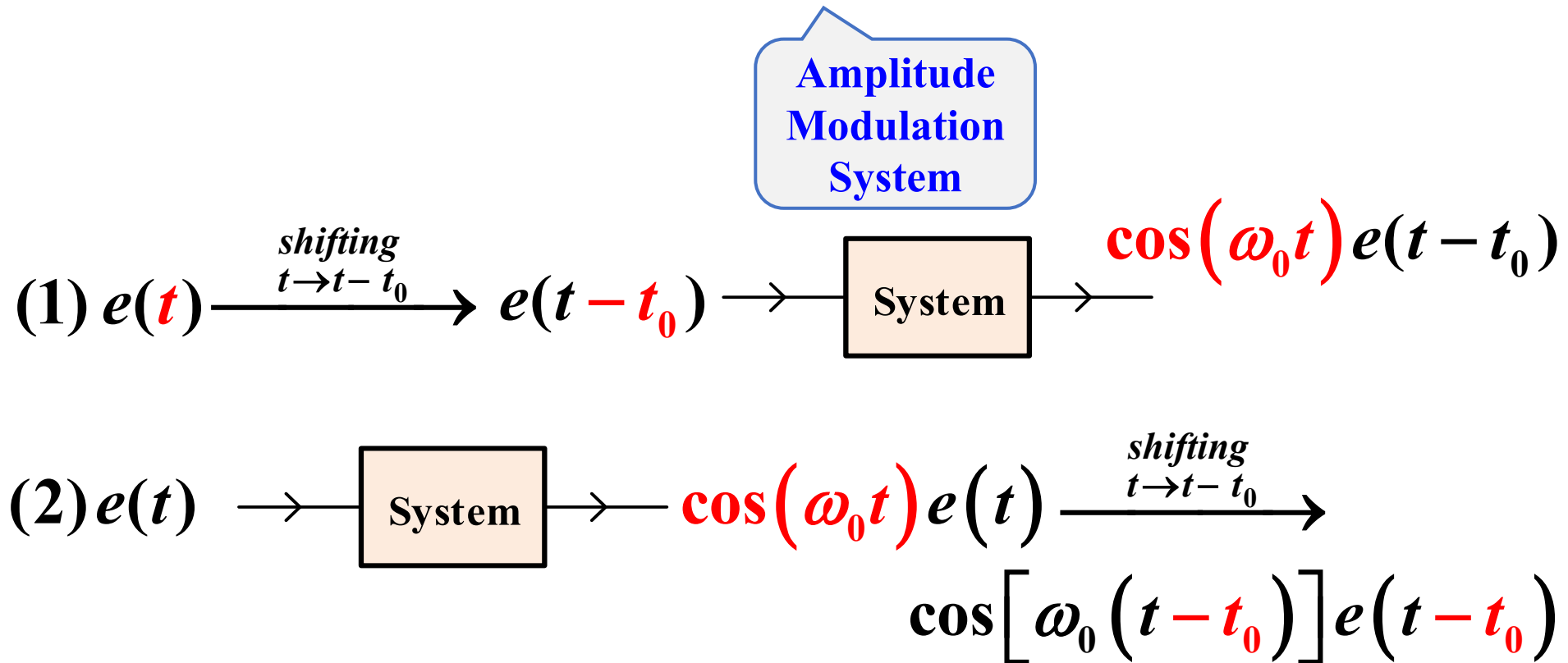
$$(1) \ e(t) \xrightarrow[\text{\textit{shifting}}]{t \rightarrow t - t_0} e(t - t_0) \rightarrow \boxed{\text{System}} \rightarrow \frac{de(t - t_0)}{dt}$$

$$(2) \ e(t) \rightarrow \boxed{\text{System}} \rightarrow r(t) = \frac{de(t)}{dt} \xrightarrow[\text{\textit{shifting}}]{t \rightarrow t - t_0} \frac{de(t - t_0)}{d(t - t_0)}$$

The results are identical ($r_1(t) = r(t - t_0)$), therefore the system is **time-invariant**

Example 3

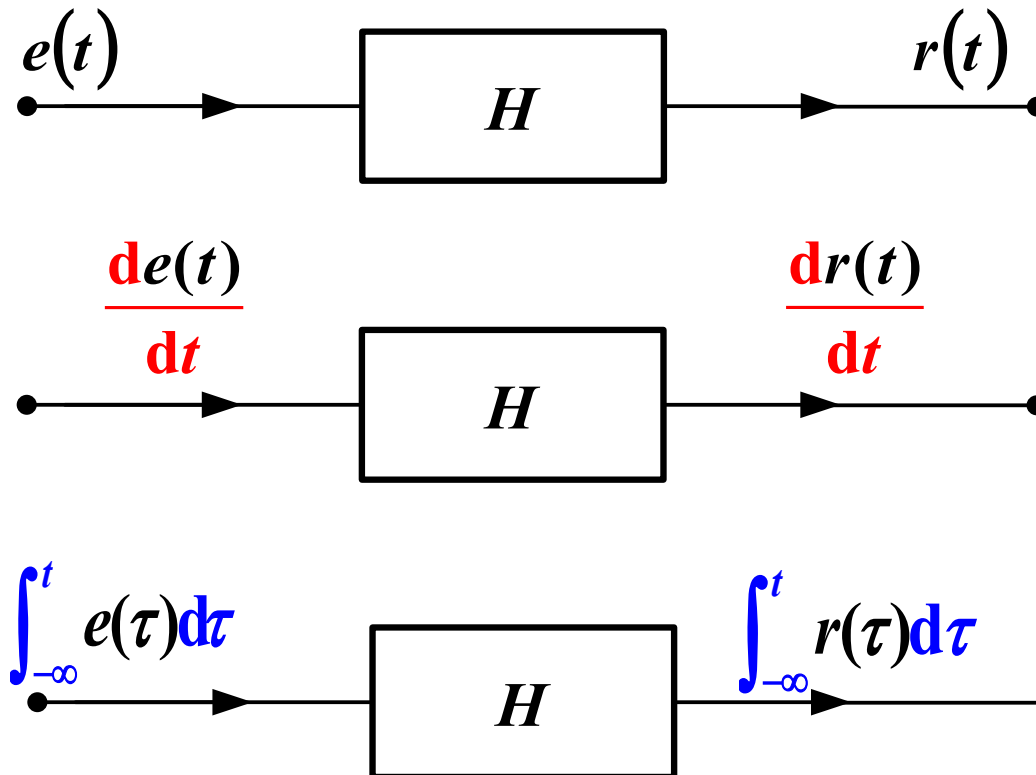
To judge whether the system $r(t) = \cos(\omega_0 t)e(t)$ is a time invariant system



Result: Different outputs $\rightarrow$ System is **time-varying**

(3) LTIS, differential characteristics, integral characteristics

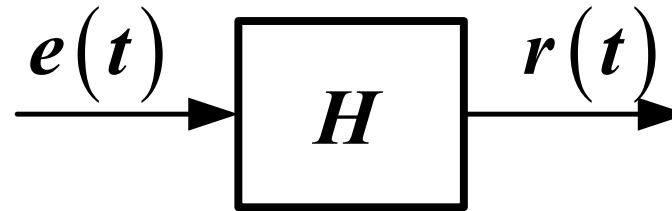
Linear and time invariant system, LTIS



7.1 System Characteristics

Proof

Using Linear and Time-Invariant Properties: The methodology can be extended to higher-order systems

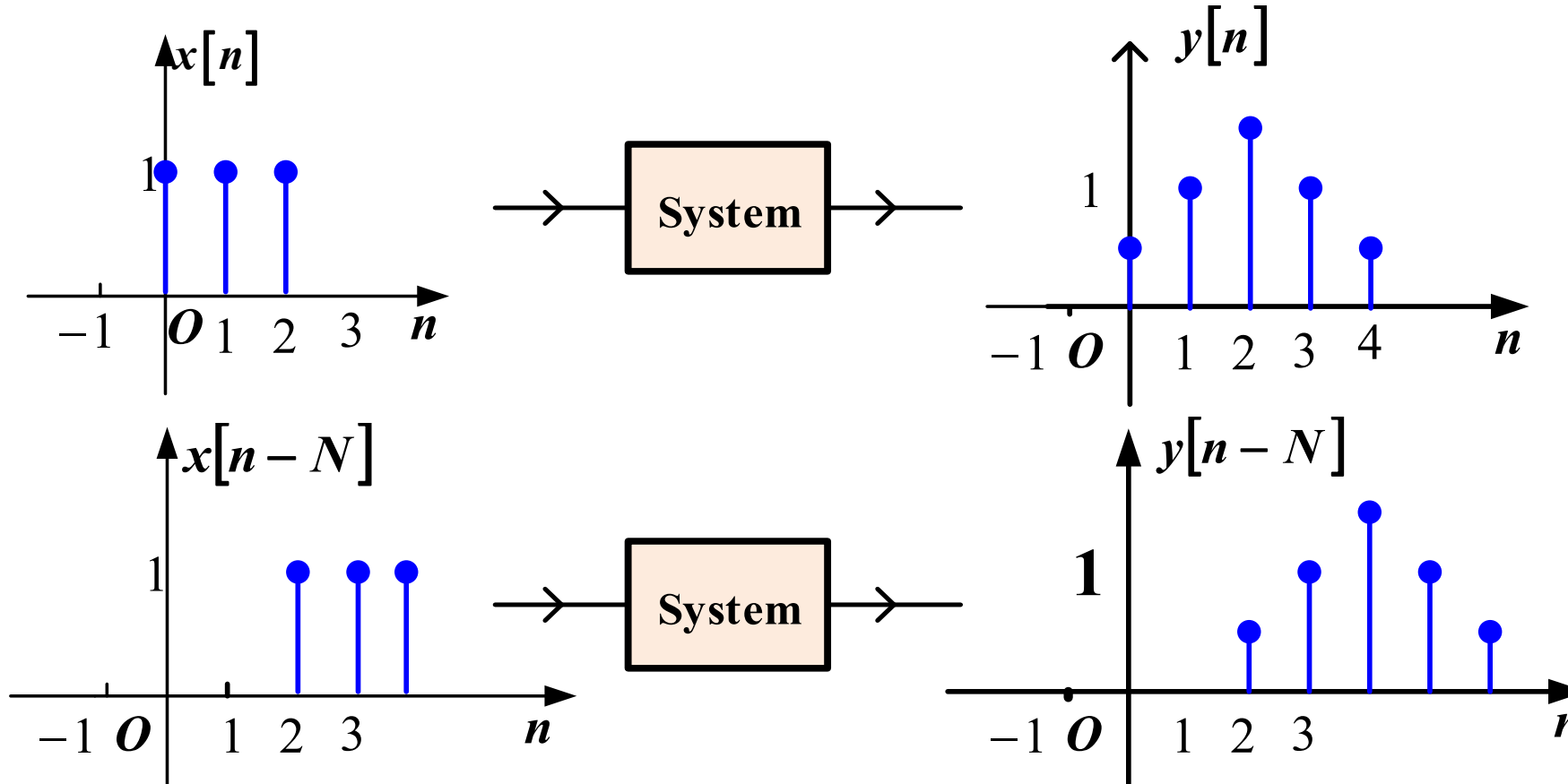


$$\frac{de(t)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{e(t) - e(t - \Delta t)}{\Delta t}$$

$$\lim_{\Delta t \rightarrow 0} \frac{r(t) - r(t - \Delta t)}{\Delta t} = \frac{dr(t)}{dt}$$

(4) Time Invariance of Discrete Systems

$$x[n] \rightarrow y[n], \quad x[n - N] \rightarrow y[n - N]$$



7. Causal Systems

Causal System

A system that produces output responses only when the input signal **excites** it

This property of the system is called **causality**

Judgment Method

Check whether the output **precedes** the input

Example 4

a. Analysis of System Causality

$$r(t) = e(t) + e(t - 2)$$

Current response = current excitation + past excitation

Therefore, the system is a Causal system

b. For the system $r(t) = e(t) + e(t + 2)$

Includes future excitation

Therefore, the system is a non-causal system

7. Causal Systems

Implementation Requirement

For a time system, implementation requires the system to be causal

Non-Causal Systems

Practical Significance: Signal compression/expansion and voice signal processing utilize non-causal system characteristics

Causal System Definition

$$e(t) = e(t)u(t)$$

Equivalent to $t < 0, e(t) = 0$

8. Stable Systems

If a system only produces **bounded outputs** for any **bounded input**, it is called a stable system in the sense of bounded input bounded output (**BIBO**)

For such a system, if the input does **not diverge**, the output will **not diverge** either

From an engineering perspective, it is crucial for a system to **remain stable** under all possible conditions.

Summary

- Continuous-Time and Discrete-Time Systems
- Memoryless (Instantaneous) Systems vs. Dynamic Systems
- Lumped-Parameter Systems and Distributed-Parameter Systems
- Invertible Systems and Non-Invertible Systems
- Linear Systems
- Time-Invariant Systems
- Causal Systems
- Stable Systems

Key Focus: Linear time invariant systems, from an implementation perspective, hope that the system is **causal** and **stable**

Main Content

- **Continuous Systems:** Differential equations, block diagrams
- **Discrete Systems:** Difference equations, block diagrams

1. Mathematical Model of Continuous Time Systems

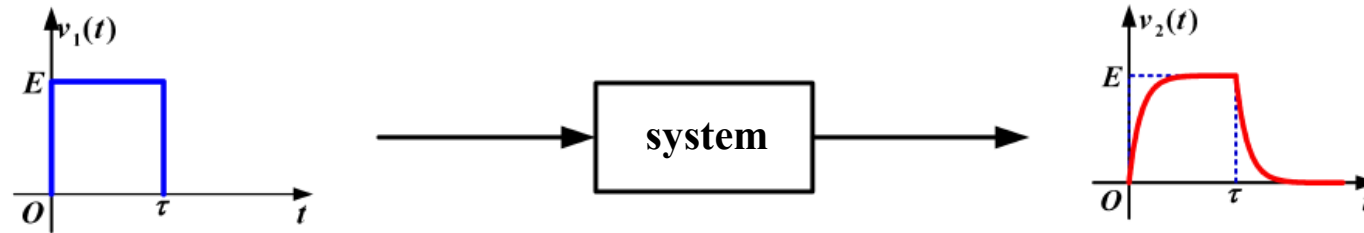
Differential equations

- Write the differential equations of the system based on the physical characteristics of the actual system
- Given the excitation conditions and initial state, calculate the response
- After mathematical analysis, we will return to physical reality

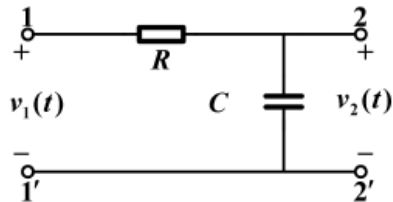
Structure diagrams, such as block diagrams and signal-flow graphs.

1. Mathematical Model of Continuous Time Systems

How to get the response of a signal through the system?



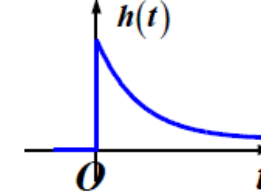
Circuit



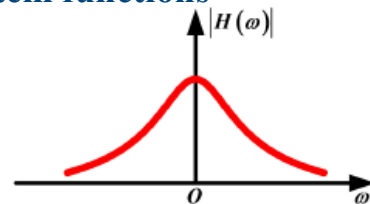
Differential equation

$$\frac{dv_2(t)}{dt} + \frac{1}{RC}v_2(t) = \frac{1}{RC}v_1(t)$$

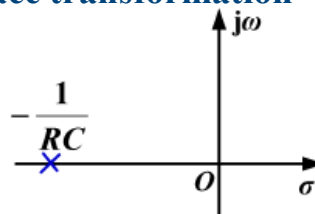
Unit impulse response



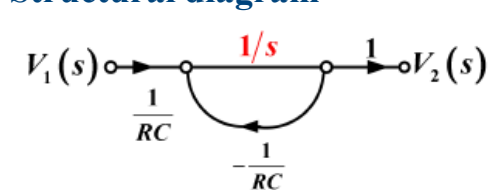
System functions



Laplace transformation



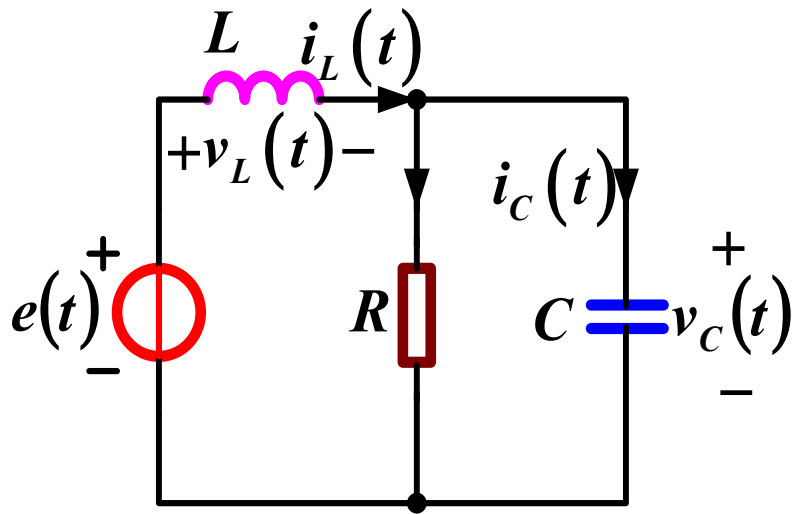
Structural diagram



7.2 Representation Method for LTIS

Example 1

Please provide the differential equations that describe this system



$e(t)$: excitation, $v_C(t)$: response

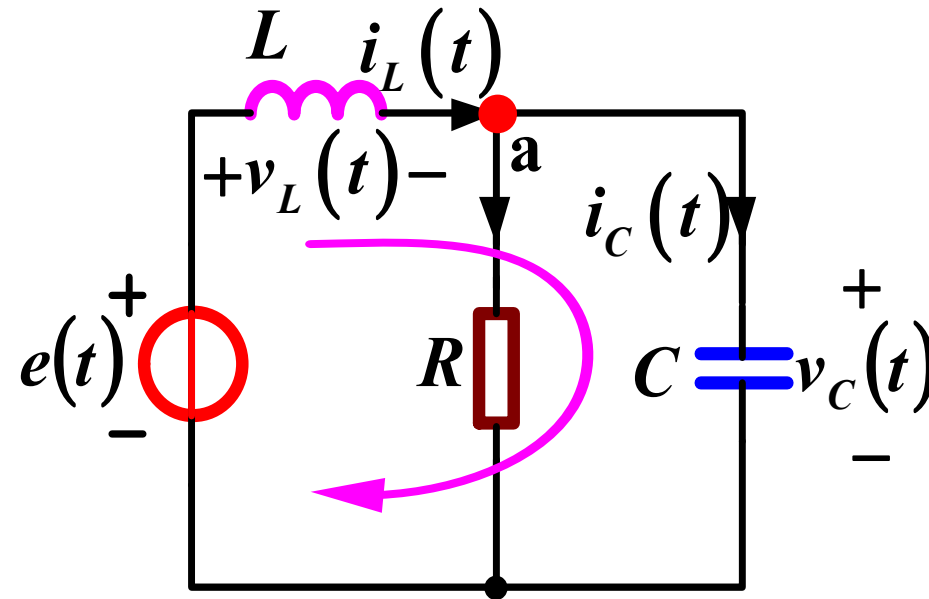
Constraint Characteristics of Components (Associated Reference Direction):

Capacitor: $i_C(t) = C \frac{dv_C(t)}{dt}$

Inductor: $v_L(t) = L \frac{di_L(t)}{dt}$

7.2 Representation Method for LTIS

Example 1



Network topology characteristics:

KVL:
$$e(t) = L \frac{d i_L(t)}{d t} + v_c(t) \quad (1)$$

KCL(Point a):
$$i_L(t) = \frac{v_c(t)}{R} + C \frac{d v_c(t)}{d t} \quad (2)$$

Example 1

$$\begin{cases} e(t) = L \frac{d i_L(t)}{dt} + v_C(t) & (1) \\ i_L(t) = \frac{v_C(t)}{R} + C \frac{d v_C(t)}{dt} & (2) \end{cases}$$

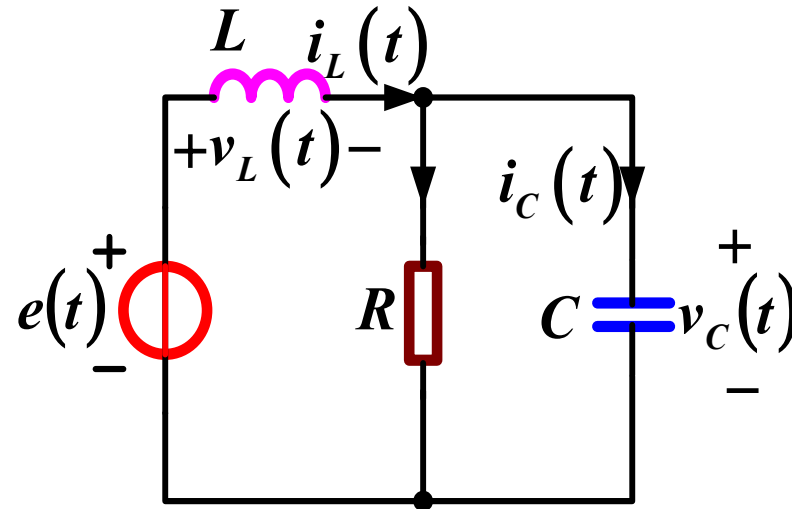
In order to **eliminate** $i_L(t)$, substitute (2) into (1) and organize it as follows

$$e(t) = L \frac{d \left[\frac{v_C(t)}{R} + C \frac{d v_C(t)}{dt} \right]}{dt} + v_C(t)$$

$$\frac{d^2 v_C(t)}{dt^2} + \frac{1}{RC} \frac{d v_C(t)}{dt} + \frac{1}{LC} v_C(t) = \frac{1}{LC} e(t)$$

7.2 Representation Method for LTIS

Example 1

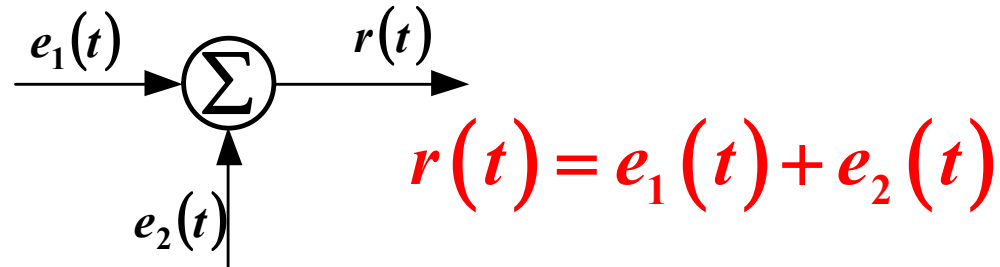


$$\frac{d^2 v_C(t)}{dt^2} + \frac{1}{RC} \frac{dv_C(t)}{dt} + \frac{1}{LC} v_C(t) = \frac{1}{LC} e(t)$$

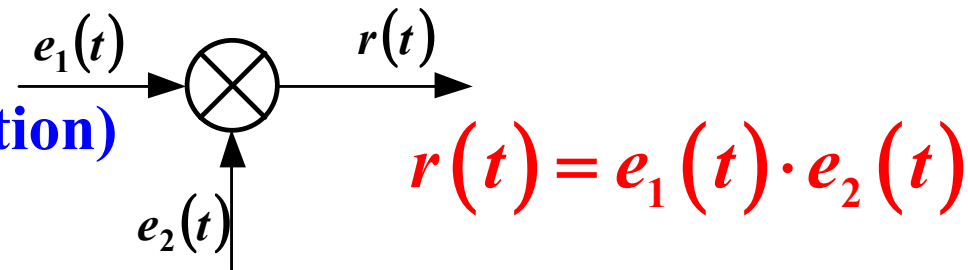
- **No** nonlinear terms (e.g., multiplication) exist in the equation $\rightarrow$ **Linear** system
- All coefficients in the equation are constants $\rightarrow$ **Time-invariant** system
- The **order** of the differential equation equals the order of the system

2. Block Diagram Representation of Continuous-Time Systems

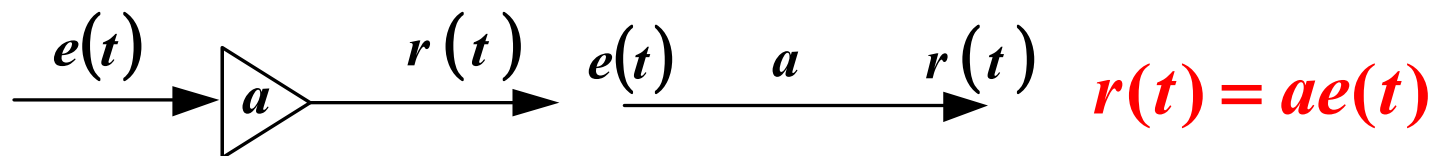
(1) Adder



(2) Multiplier
(nonlinear operation)

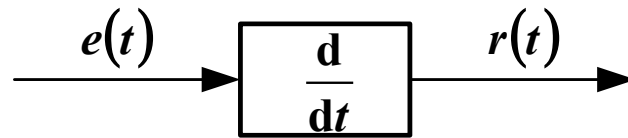


(3) Scalar multiplier (coefficient multiplier, scaler)



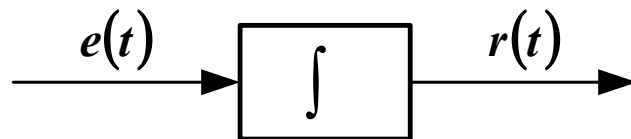
2. Block Diagram Representation of Continuous-Time Systems

(4) Differentiator



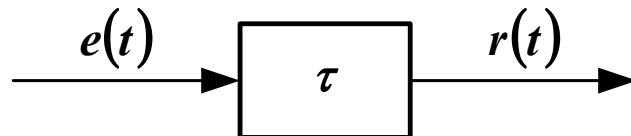
$$r(t) = \frac{de(t)}{dt}$$

(5) Integrator



$$r(t) = \int_{-\infty}^t e(\tau) d\tau$$

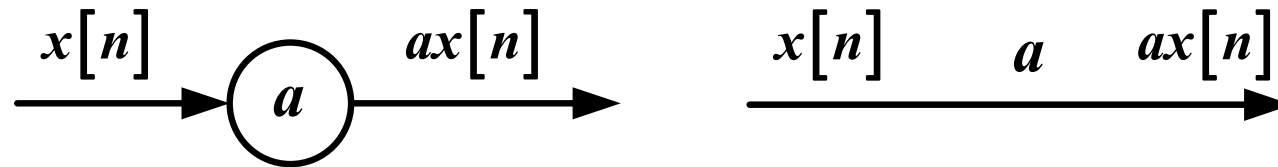
(6) Time-delay element



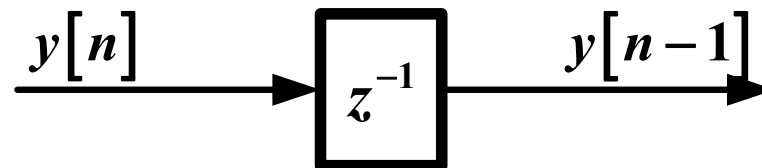
$$r(t) = e(t - \tau)$$

3. Representation of Discrete-Time Systems

Scalar Multiplier



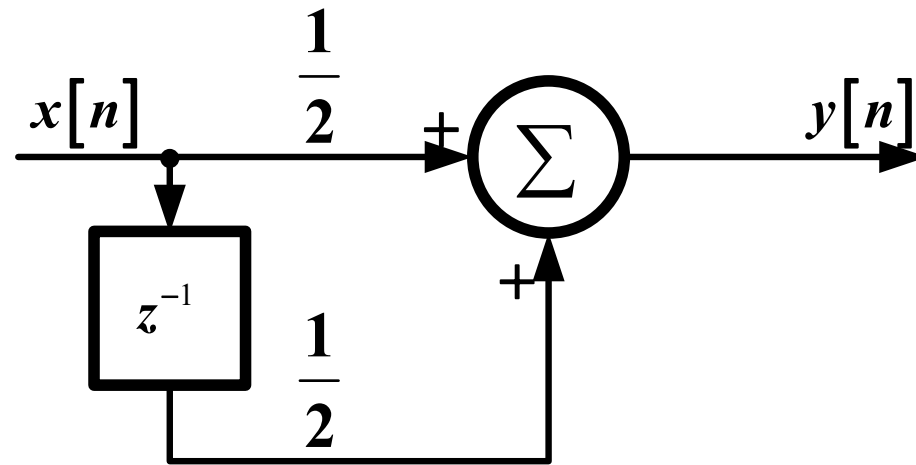
Unit Delay Element



A unit delay is essentially a shift register that pushes out the previous value while shifting in the new one

7.2 Representation Method for LTIS

Example 2

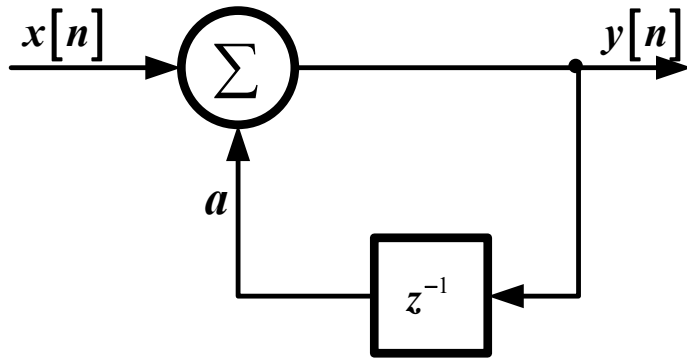


$$y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$$

System Without Feedback Terms

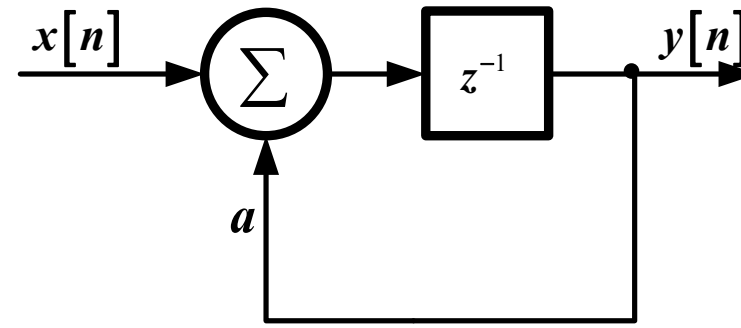
7.2 Representation Method for LTIS

Example 3



$$y[n] = x[n] + ay[n-1]$$

First order backward difference equation



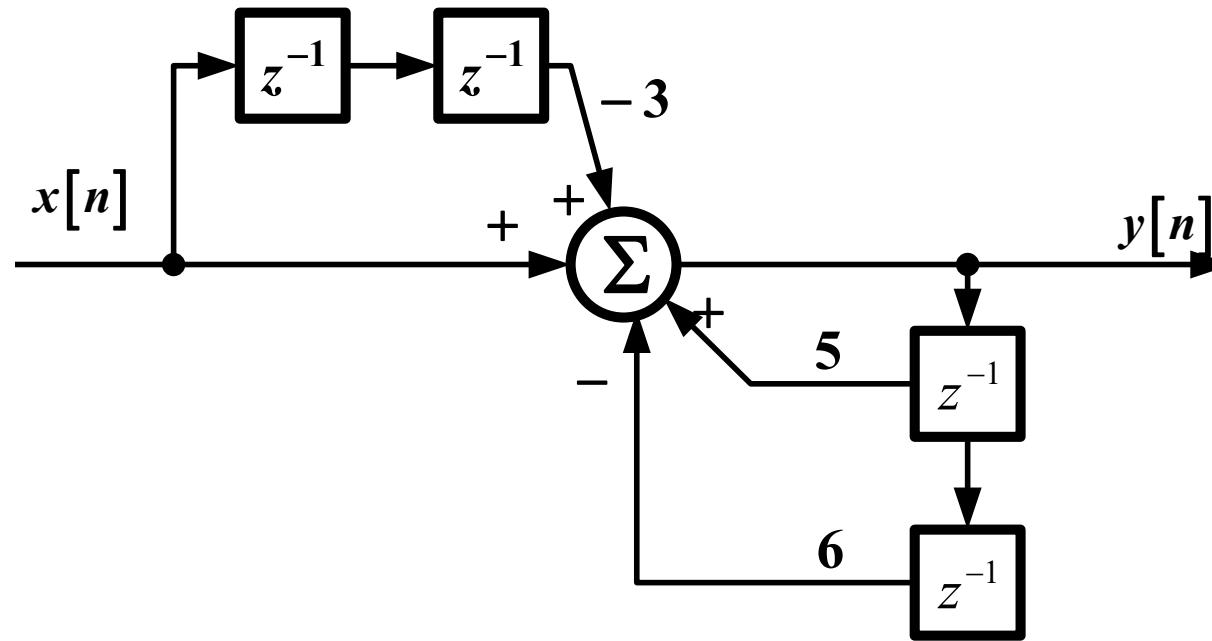
$$y[n+1] = x[n] + ay[n]$$

First-order forward difference equation

We use a backward difference equation

7.2 Representation Method for LTIS

Example 4



$$y[n] = x[n] - 3x[n-2] + 5y[n-1] - 6y[n-2]$$

$$\textcolor{red}{y}[n] - 5\textcolor{red}{y}[n-1] + 6\textcolor{red}{y}[n-2] = \textcolor{blue}{x}[n] - 3\textcolor{blue}{x}[n-2]$$

A **second-order system** can be represented through multiple structural diagrams, all requiring at least **two delay elements** for implementation

3. Representation of Discrete-Time Systems

General Form
$$\sum_{k=0}^N a_k y[n-k] = \sum_{r=0}^M b_r x[n-r]$$

Order of a Difference Equation: The difference between the highest and lowest argument values of the dependent variable in the equation

Summary

- **Continuous Systems:** Differential equations, block diagrams
- **Discrete Systems:** Difference equations, block diagrams

Main Content

- Overview of System Analysis Methods
- Two Approaches to Establish System Models
- Mathematical Model Solution Methods

1. Overview of System Analysis Methods

Research Objectives:

To investigate fundamental analysis methods for **linear time-invariant (LTI)** systems, with applications to signal transmission and processing in communication systems

System Analysis Approach:

(1). Mathematical Modeling

Develop equations characterizing the system dynamics

(2) Response Analysis

Solve for system outputs under given inputs

2. Two Approaches to Establish System Models

Input-Output Description Method:

1. Focuses solely on the relationship between excitation and response, disregarding internal system variables
2. Applicable to single-input/single-output (SISO) systems
3. Formulates a single **n th-order ordinary differential equation**

State-Variable Analysis Method:

1. Capable of determining both system response and internal variables (e.g., capacitor voltages, inductor currents)
2. Suitable for multiple-input/multiple-output (MIMO) systems
3. Establishes a set of coupled **first-order differential equations**

3. Mathematical Model Solution Methods

1. Time Domain

Classical method for solving { Continuous Systems: Differential Equations
Discrete Systems: Difference Equations

Convolutional integral (or convolutional sum) method

2. Transform-Domain Analysis

- Fourier transform (FT)
- Laplace transform (LT)
- Z transform (ZT)
- Fourier Transform of Discrete Time Signals DFT

4. Stable Systems

Input-Output Description $\rightarrow$ State-Variable Description

Time-Domain Analysis $\rightarrow$ Transform-Domain Analysis

Continuous Systems $\rightarrow$ Discrete Systems



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for listening !

